

Abstract No.

Rewiring glioblastoma immunity: the role of EGFL7-Integrin $\beta 2$ axis

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Background: Glioblastoma is the most aggressive form of malignant brain cancer, characterized by an immunosuppressive microenvironment and immune evasion. Despite the success of immune checkpoint inhibitors in other cancers, immunotherapies such as anti-PD1 have shown limited efficacy in glioblastoma, underscoring the need to identify tumor-intrinsic mechanisms that sustain this immunosuppressive microenvironment and to develop more effective therapeutic strategies targeting them. Previously, the secreted factor **epidermal growth factor-like protein 7 (EGFL7)** has been shown to promote brain tumor growth by affecting the glioblastoma microenvironment (GME). However, its impact on the immune system remains enigmatic.

Material and Methods: Using in vivo glioma models, the role of EGFL7 in shaping the immune landscape in glioblastoma was studied through **single cell RNA-sequencing**, spatial transcriptomics, flow cytometry and mass spectrometry, and analysis of **human patient data** to corroborate our study.

Results: Single-cell transcriptomic profiling of immune cells derived from glioblastoma revealed that EGFL7 promotes an immunosuppressive GME, characterized by **enhanced T cell exhaustion** and **polarization of macrophages** towards a protumorigenic state. To uncover the underlying mechanism driving this immunomodulation, proteomic profiling of EGFL7's interactome in glioma revealed its **interaction with integrin $\beta 2$ (ITGB2)**, an immune cell surface receptor involved in cell adhesion and migration. Building on this finding, genetic perturbation of the EGFL7-ITGB2 interaction attenuated immunosuppression by limiting T cell exhaustion and reducing polarization of macrophages towards a pro-tumorigenic state, along with prolonging the survival of glioblastoma-bearing mice.

Remarkably, a combinatorial regimen of anti-EGFL7 and checkpoint inhibitor anti-PD1 improved the efficacy of the drug, which by itself has not improved glioma patient survival so far.

Conclusion: Our study provides unequivocal evidence that EGFL7 mediates immune evasion in glioblastoma and has great potential to serve as an **add-on drug target** to improve the efficacy of immunotherapies which are not functional in glioblastoma patients so far.



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